

Automatic Speech Recognition in Bengali Low-Resource Language Dialects: A Mini-Review of Applications, Challenges, and Emerging Trends

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Abstract

Bengali (Bangla) is one of the most widely spoken languages in the world, but speech technology for its many regional dialects is still weak and under developed. Automatic speech recognition for these dialects is important for real uses such as subtitles, voice search and access to services in Bangladesh, yet most systems are trained only on standard Bangla. In recent years self supervised encoders and large multilingual models like wav2vec 2.0, Wav2Vec BERT, WavLM and Whisper have made it easier to build ASR for low resource languages and have reached good results on standard Bangla. At the same time, new work on Bengali shows that these models still make many errors when the speaker uses a local dialect or when the recording is noisy. This mini review brings together the main results from that work. It looks at how different model families behave on Bangla, how a WavLM based framework can be adapted to dialect and noise, and what the Ben 10 dialect corpus reveals about the limits of current systems. From this, the paper highlights what is already possible for Bengali dialect ASR, what problems remain open and what research directions are most promising for building practical systems for Bangladeshi local dialects.

1 Introduction

Bengali is the primary language of Bangladesh and the second most widely spoken language in India, with hundreds of millions of speakers overall. From the perspective of automatic speech recognition (ASR), however, it is still treated as a low-resource language: the amount of freely available, high-quality, transcribed speech is far smaller than for English or Mandarin, and most public datasets target a relatively narrow standard variety. This mismatch between demographic importance and technological support becomes more severe when considering regional Bengali dialects, especially those spoken in rural Bangladesh, where phonology, prosody and lexicon diverge substantially from Standard Colloquial Bangla (SCB). In such settings, state-of-the-art ASR systems trained mostly on SCB may exhibit very high error rates and thus be unusable in practice.

ASR technology itself has evolved rapidly over the past decade. Early end-to-end systems such as Deep Speech replaced the conventional HMM-GMM pipeline with a deep recurrent neural network trained with the Connectionist Temporal Classification (CTC) loss, directly mapping spectrograms to character sequences and achieving competitive performance on English benchmarks under noisy conditions [1]. However, these systems still depended on thousands of hours of transcribed audio, which is impractical for most of the world’s languages.

Two more recent developments are especially relevant for low-resource scenarios. First, self-supervised learning (SSL) has made it possible to learn powerful speech representations from large amounts of unlabelled audio. wav2vec 2.0 masks latent speech representations and trains a Transformer encoder with a contrastive objective; when the resulting encoder is fine-tuned on labelled data, it achieves state-of-the-art word error rates (WER) on LibriSpeech and remains competitive even when only minutes of transcribed speech are available [2]. WavLM extends this approach by adding a masked speech denoising pre-training objective and a larger, more diverse pre-training corpus, yielding representations that are robust to reverberation and additive noise and can support “full stack” speech processing tasks [3]. Second, large multilingual “foundation” ASR models trained on web-scale weak supervision have emerged. Whisper is a prominent example: its Transformer encoder-decoder is trained on roughly 680 000 hours of multilingual audio-text pairs and can perform multilingual ASR, speech translation and language identification without task-specific fine-tuning on many benchmarks [4].

These models are attractive for low-resource languages because they can leverage unlabelled audio or multilingual transfer instead of relying solely on expensive manual transcription. However, whether they generalise to Bengali regional dialects is an empirical question. Conventional work on Bangla ASR has largely focused on standard varieties; dialects are often left to future work or treated as a minor fine-tuning issue. Recent research begins to challenge this assumption through three complementary strands. First, Ridoy *et al.* compare Whisper and Wav2Vec-BERT on Bangla ASR using only publicly available corpora and show that the self-supervised encoder outperforms Whisper in both accuracy and efficiency under low-resource conditions [5]. Second, Biswas *et al.* introduce a WavLM-based unified denoising and adaptation framework that explicitly targets dialectal Bengali under noisy conditions and demonstrates substantial gains over wav2vec 2.0 and Whisper [6]. Third, Tashwar *et al.* present the Ben-10 corpus, a 78-hour speech-to-text dataset covering ten regional dialects in Bangladesh, and show that foundation models struggle heavily on this out-of-distribution benchmark both before and after fine-tuning [7].

This mini-review focuses on these three contributions and the underlying model families they rely on. Our aim is not to survey all work on Bangla ASR, but to provide a concise, research-style overview of applications and emerging trends in ASR for low-resource Bengali dialects, with an eye towards Bangladeshi local varieties. The remainder of the paper is organised as follows. Section 2 reviews the state of the art in Bengali ASR for low-resource dialects, centred on the Whisper vs. Wav2Vec-BERT comparison, the WavLM-based dialectal framework and the Ben-10 corpus. Tables 1 and 2 at the end of Section 2 give an overview of the main papers and datasets used in this review. Section 3 discusses applications and emerging trends suggested by these works and uses Table 3 to summarise common problems and solution directions. Section 4 concludes with open problems and research directions for future work on Bangladeshi dialect ASR.

2 State of the Art in Bengali ASR for Low-Resource Dialects

Work on Bengali dialectal ASR sits at the intersection of three developments: self-supervised acoustic encoders, large multilingual foundation models and dialect-specific corpora. This section concentrates on three recent Bengali-centric studies and situates them with respect to their underlying architectures.

Ridoy *et al.* investigate the adaptability of ASR models on low-resource Bangla in a comparative study of Whisper and Wav2Vec-BERT [5]. The authors use the Bangla subset of Mozilla Common Voice 17 and OpenSLR SLR63 as training corpora; together these provide roughly 90–100 hours of transcribed standard Bangla from diverse speakers. They fine-tune Whisper (Small and Large-V2) and a Wav2Vec-BERT configuration while varying dataset size and number of epochs. The main result is that Wav2Vec-BERT achieves lower WER and character error rate (CER) than Whisper across the tested conditions while also requiring fewer computational resources. For example, with 70 000 training samples and eight epochs, Wav2Vec-BERT attains a WER of around 14.4% and a CER of around 2.7%, whereas Whisper Small and Large-V2 yield substantially higher error rates on comparable setups. Excessive fine-tuning epochs lead to over-fitting, highlighting the sensitivity of low-resource training. Overall, this study supports the view that self-supervised encoders are strong candidates for Bangla ASR and may be particularly suitable as backbones for dialectal adaptation.

While Ridoy *et al.* focus on standard Bangla, Biswas *et al.* explicitly address dialectal and noisy Bengali speech in their unified denoising and adaptation framework for self-supervised Bengali dialectal ASR [6]. They begin by noting that Bengali ASR is challenged by two intertwined factors: extensive dialectal diversity and frequent acoustic noise in real-world recordings, from urban traffic to crowded markets. Standard SSL models such as wav2vec 2.0 and large multilingual models such as Whisper, although powerful, are not explicitly trained to handle this combination. Their proposed framework builds on WavLM, an SSL model pre-trained with a masked speech denoising objective on a large, heterogeneous English speech corpus [3]. WavLM’s pre-training corrupts input audio with noise and reverberation and trains the model to reconstruct clean features, yielding representations that are inherently robust to acoustic distortions.

The framework performs multi-stage fine-tuning. In the first stage, WavLM-Large is adapted to standard Bengali using Common Voice 17 Bangla and OpenSLR SLR63, establishing a strong linguistic base. In the second stage, the model is further fine-tuned on dialectal Bengali speech under simulated noise conditions. Dialectal data are drawn from multiple varieties, including e.g. Chittagonian and Sylheti, and on-the-fly augmentation mixes these utterances with noise from MUSAN at signal-to-noise ratios (SNR) ranging from clean to 0 dB. The authors evaluate WER across dialects and SNR levels and compare against two strong baselines: a fine-tuned wav2vec 2.0 model and a Whisper-based system. Across conditions, their WavLM-based approach significantly outperforms both baselines, with especially large gains at

low SNR. At 0 dB, relative WER reductions can exceed 40 % compared to wav2vec 2.0 and nearly 50 % compared to Whisper [6]. These results demonstrate that noise-aware SSL pre-training plus dialect-focused fine-tuning can yield tangible improvements for Bengali dialectal ASR.

Tashwar *et al.* tackle the problem from a complementary angle by constructing a dedicated dialect corpus and out-of-distribution benchmark [7]. Their Ben-10 dataset comprises more than 78 hours of annotated speech-to-text data from 394 speakers across ten Bangladeshi regions, each contributing more than five hours of audio. The 16 690 clips (average duration ≈ 16.6 s) span 155 topics and emphasise spontaneous, conversational speech recorded in natural, often noisy environments. Linguistic analysis using acoustic features and dimensionality reduction shows clear distributional differences between Ben-10 and canonical Bengali speech corpora, confirming its status as an out-of-distribution resource.

With Ben-10 in hand, the authors evaluate a range of ASR systems: wav2vec 2.0 fine-tuned on SCB, the same architecture fine-tuned on Ben-10, a large Conformer-based “Hishab” model trained on approximately 20 000 hours of pseudo-labelled Bengali audio, Whisper-large-v3, Google’s commercial ASR and competition-winning systems. Across these systems, zero-shot performance on Ben-10 is consistently poor, with high WER and CER for all dialects. Fine-tuning on Ben-10 improves the results but does not close the gap to performance on canonical Bengali; large differences between dialects remain [7]. The authors conclude that contemporary speech foundation models are not yet generalised enough to capture regional dialect features in low-resource languages and that dialect-specific corpora and model training are essential.

Taken together, these three studies define the current state of the art for Bengali low-resource dialect ASR. Wav2Vec-BERT appears more data- and compute-efficient than Whisper on standard Bangla [5]. WavLM, with denoising-aware pre-training and multi-stage adaptation, offers improved robustness for dialectal and noisy speech [6, 3]. Ben-10 provides a challenging testbed showing that even large, carefully trained systems still struggle with regional Bangladeshi dialects, especially in spontaneous conversations [7].

To give a compact overview of the literature and resources discussed above, Table 1 summarises the main reviewed papers, and Table 2 lists the key Bengali-related datasets.

Table 1: Summary of the reviewed papers and how they relate to Bengali dialect speech technology.

Paper	Main focus	Data / languages	Main contribution and relevance
Hannun et al. (2014) [1]	End-to-end ASR (Deep Speech)	English (LibriSpeech, Switchboard)	Introduces Deep Speech, a large-scale end-to-end RNN + CTC ASR system. Important as an early end-to-end baseline and motivation for moving away from traditional GMM-HMM pipelines towards neural ASR that can later be adapted to low-resource languages.
Baevski et al. (2020) [2]	Self-supervised ASR (wav2vec 2.0)	Mainly English (LibriSpeech)	Proposes wav2vec 2.0, a self-supervised framework for learning speech representations from raw audio. Shows that strong ASR can be trained with only small amounts of labelled data. Provides the main SSL backbone concept used in later Bangla and dialectal work.
Chen et al. (2022) [3]	Noise-robust SSL (WavLM)	English (large multi-corpus mix)	Introduces WavLM, which uses masked speech denoising pre-training for noise-robust speech representations. Forms the technical basis for the Bengali dialect ASR framework that targets noisy real-world recordings.
Radford et al. (2023) [4]	Foundation ASR (Whisper)	Multilingual, many tasks	Presents Whisper, a large-scale multilingual and multitask encoder-decoder ASR model trained on web-scale weak supervision. Serves as a strong baseline for Bangla and as a foundation model whose limitations on dialectal Bengali are highlighted by later work.
Ridoy et al. (2025) [5]	Bangla ASR model comparison	Bangla (Common Voice v17, OpenSLR SLR63)	Compares Whisper and Wav2Vec-BERT for Bangla ASR. Shows that Wav2Vec-BERT can outperform Whisper in both accuracy and efficiency under low-resource conditions. Motivates using self-supervised encoders as backbones for Bengali dialect ASR.

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Paper	Main focus	Data / languages	Main contribution and relevance
Biswas et al. (2025) [6]	Bengali dialect ASR with noise	Bengali (standard + dialectal OOD speech)	Proposes a unified denoising and adaptation framework based on WavLM with multi-stage fine-tuning: first on standard Bangla, then on dialectal and noisy speech. Evaluates on dialects such as Chittagonian and Sylheti under controlled noise levels and shows clear gains over wav2vec 2.0 and Whisper.
Tashwar Dipto et al. (2025) [7]	Dialect corpus and benchmarking (Ben-10)	Bangladeshi regional dialects	Introduces Ben-10 (over 78 hours, 10 regions, 394 speakers) and benchmarks several ASR foundation models. Shows that these models struggle on dialect speech and that dialect-specific training is needed. Provides a key out-of-distribution testbed for Bangladeshi dialect ASR.
de Zuazo et al. (2025) [8]	Whisper + LM integration (Whisper-LM)	Multiple low-resource languages	Shows that integrating external language models with fine-tuned Whisper improves ASR performance in low-resource languages. Explores both n-gram and neural LMs. Relevant for improving decoding of rare and dialect-specific words in Bengali ASR.
Li & Niehues (2025) [9]	Contextual learning and LM integration	Endangered low-resource languages (5)	Studies tokenization choices and language-model integration in ASR for endangered languages. Finds that fine-grained tokenization and carefully tuned LM integration can improve performance. Provides guidance for LM use in Bengali dialect ASR.
Dhahbi et al. (2025) [10]	Synthetic speech and augmentation for low-resource ASR	Bengali, Hindi, Turkish, Urdu	Proposes an end-to-end ASR system that uses FastSpeech-based synthetic speech and spectrogram augmentation. Shows that joint training on real, synthetic and augmented data reduces WER/CER, including for Bengali. Informs data augmentation strategies for Bangladeshi dialects.
Hafiz Imam et al. (2025) [11]	Low-resource ASR survey (African)	African low-resource languages	Surveys challenges and future directions for African low-resource ASR: data scarcity, dialect diversity, SSL and multilingual transfer, and ethical issues. Provides a broader context and fairness perspective that is relevant when designing inclusive Bengali dialect ASR.
Yang et al. (2025) [12]	Large-scale corpus pipeline (GigaSpeech 2)	Thai, Indonesian, Vietnamese	Presents GigaSpeech 2 (about 30,000 hours) and an automated crawling, transcription and refinement pipeline using Whisper and forced alignment. Offers a scalable recipe that can inspire semi-automatic collection of Bangladeshi dialect speech.

Table 2: Datasets and resources referenced in the Bengali-related papers. Sizes are approximate and reported as stated in the cited papers.

Dataset / resource	What it covers	Size	Source paper(s)	Notes for Bangladesh dialect work
Common Voice Bangla v17	Mostly standard Bangla read speech	about 54 hours verified	[5, 6]	Good for base adaptation and speaker diversity, but has limited explicit dialect coverage. Useful as a first-stage training set for Bengali ASR models.
OpenSLR Bangla (SLR63)	Bangla read speech with many prompts	about 40 hours	[5, 6]	Adds lexical and prompt diversity; still focused on standard forms. Complements Common Voice Bangla for standard-language adaptation.
Ben-10	10 Bangladesh regions, mostly spontaneous speech	78+ hours; 16,690 clips	[7]	Dialect-focused benchmark with strong regional diversity. Shows a large out-of-distribution shift compared to standard Bangla and is essential for testing Bangladeshi dialect ASR.
OOD-Speech (dialect set)	Challenging out-of-distribution Bangla dialect speech	about 20 hours (reported)	[6]	Used for dialect specialisation and evaluation in the WavLM framework. Includes dialects such as Chittagonian and Sylheti; helpful for fine-tuning dialect-aware models.
IARPA Babel subsets (used)	Conversational telephone speech	Bengali about 62 hours; Turkish about 10 hours (reported)	[10]	Shows that conversational telephone speech differs from read speech and can be used to simulate more realistic call-centre style conditions for Bengali ASR.
GigaSpeech 2	YouTube audio-only corpus (multi-domain)	about 30,000 hours	[12]	Not Bengali, but demonstrates an automated pipeline for large-scale crawling, transcription and refinement in low-resource languages. Can inspire similar pipelines for Bangladeshi dialect audio from online platforms.

3 Applications and Emerging Trends

Although the reviewed work is recent, it already suggests several application-oriented trends for ASR in Bangladeshi local dialects.

A first trend concerns model choice in low-resource settings. Evidence from Ridoy *et al.* indicates that, when training data and compute are constrained, self-supervised encoders such as Wav2Vec-BERT can outperform a powerful foundation model like Whisper on Bangla [5]. Combined with the WavLM results, this points towards a design pattern in which a single SSL backbone is adapted to standard Bangla and then specialised for dialects. In practical systems deployed in Bangladesh—such as voice-based interfaces for government services or local call-centre automation—this design may be preferable to using larger Whisper variants, which are more resource-hungry at inference time and may be harder to adapt effectively to dialects.

A second trend relates to noise robustness and realistic deployment environments. Biswas *et al.* explicitly target scenarios where speakers interact with ASR in noisy streets, buses or markets, conditions common in Bangladeshi cities and rural centres. Their experiments show that models not explicitly trained to handle such noise (e.g. standard wav2vec 2.0 or Whisper fine-tuned only on clean data) degrade severely at low SNR, whereas WavLM fine-tuned with simulated noise maintains substantially lower WER [6, 3]. For applications such as automatic captioning of regional news, voice search in mobile apps or transcription of telephone-quality audio, these results emphasise that explicit noise modelling is essential: achieving robustness requires both noise-aware pre-training and targeted augmentation during adaptation.

A third emerging trend is the recognition that dialectal corpora are central infrastructure, not merely evaluation add-ons. Ben-10 goes beyond treating dialect examples as a held-out test set; it is built to enable both model training and systematic linguistic analysis across regions [7]. For Bangladeshi applications, this suggests that investments into community-driven data collection—recording speakers in different districts with metadata about region, age and gender—are foundational for any serious dialect-aware ASR effort. Without such resources, error patterns specific to, for example, Chattogram or Sylhet will remain hidden behind aggregate WER numbers dominated by SCB.

Fourth, these works hint at multi-stage and domain-aware fine-tuning as a general recipe. Instead of training a single model on a mixed dataset, the WavLM framework separates adaptation into a standard-Bangla stage and a dialect/noise stage [6]. This structured approach allows the model to first stabilise on canonical phonetic and orthographic patterns and then specialise to dialectal variants and acoustic conditions. For Bangladeshi local dialects, similar multi-stage strategies could be extended further, for example by adding a code-switching-focused stage for urban Bangla-English speech or a domain-specific stage for sectors such as healthcare and agriculture. Lessons from low-resource and endangered-language ASR suggest that language-model integration and tokenisation choices will also need to be tuned specifically for Bengali dialects rather than copied from high-resource languages [8, 9, 10, 11].

A fifth trend concerns the limits of zero-shot generalisation in current foundation models. Whisper is often promoted as a robust, “ready-out-of-the-box” system, but the Ben-10 results show that its zero-shot performance on regional Bangladeshi dialects is far from acceptable [7, 4]. Similarly, large models like Hishab Conformer, trained on tens of thousands of hours of pseudo-labelled Bengali audio, do not automatically transfer to dialectal speech. This suggests that foundation models cannot substitute for dialect-specific data and adaptation, at least in the current generation.

Finally, there is an implicit but important fairness dimension. While the Bengali papers do not always frame their findings in fairness terms, the dialect-wise performance differences reported in Ben-10 and the strong gains seen when dialect-focused training is applied indicate that current systems risk systematically excluding speakers from under-represented regions [7, 11]. For Bangladeshi applications, this has social implications: a voice-based government service that recognises Dhaka-centric speech well but fails on rural dialects effectively deepens existing inequities. Integrating fairness-aware evaluation—such as reporting WER per dialect and per demographic group—into future Bengali ASR studies therefore appears to be an emerging, and necessary, trend.

Table 3: Problem patterns and solution directions supported by the reviewed papers.

Problem pattern in Bangladesh dialect ASR	Evidence from reviewed papers	Solution directions suggested by the papers
Dialect speech is out-of-distribution for standard-trained ASR	Ben-10 shows a strong distribution shift and poor performance of foundation models and other ASR systems on dialect speech, even after fine-tuning on standard Bangla [7].	Build dialect corpora and train dialect-aware models. Use dialect-specific training and evaluation splits and report metrics per dialect rather than a single global WER [7].
Noise and channel mismatch	Bengali dialect ASR is harmed by real-world noise and channel conditions. The WavLM-based framework evaluates performance down to 0 dB SNR and shows large degradation for models not trained with noise [6].	Use noise-robust self-supervised backbones (such as WavLM) and apply noise augmentation during dialect adaptation. Treat noisy and reverberant conditions as a core part of training rather than an afterthought [6, 3].
Compute limits and scaling limits	Whisper can require high compute, while Wav2Vec-BERT can be more efficient. The Bangla comparison study also observes diminishing returns when only adding more data and epochs without careful tuning [5].	Pick a model that matches available hardware. Tune fine-tuning length and batch sizes carefully. Focus on better data, better pre-training and better decoding strategies instead of only scaling model size [5, 2].
Weak decoding for rare and dialect words	Low-resource settings have many rare words and dialect forms. Work on endangered languages and Whisper-LM shows that tokenization and language-model integration strongly affect final ASR quality [9, 8].	Use fine-grained tokenization (e.g., characters or small subwords) and integrate external language models (n-gram or neural) into decoding. Tune LM weights and penalties for Bengali specifically rather than copying settings from other languages [9, 8].
Unequal performance across speaker groups and regions	Surveys on low-resource languages report that ASR systems often perform worse for certain regions and demographic groups, and that dialects are systematically under-served [11]. Results on Ben-10 also show large differences between dialects [7].	Evaluate models by dialect and by demographic subgroup where possible. Treat inclusion and fairness as design requirements. Collect additional data for under-represented dialects and consider re-weighting or targeted adaptation to reduce gaps [11, 7].

4 Conclusion and Future Work

The recent literature on Bengali low-resource dialect ASR paints a nuanced picture. On the one hand, advances in self-supervised learning and weakly supervised foundation models have dramatically improved the feasibility of building ASR systems for Bangla with limited labelled data. wav2vec 2.0 and its successors show that high-quality acoustic representations can be learned from large unlabelled corpora and adapted efficiently; Wav2Vec-BERT, in particular, has been shown to outperform Whisper on standard Bangla under realistic resource constraints [2, 5]. On the other hand, the introduction of Ben-10 and the WavLM-based dialectal framework demonstrate that regional dialects and noisy conditions remain open challenges. Foundation ASR models, even when large and carefully trained, do not generalise reliably to Bangladeshi local dialects in zero-shot mode, and dialect-specific fine-tuning, while helpful, does not fully close the gap [6, 7].

Looking ahead, several directions appear particularly promising for research focused on automatic speech recognition in Bengali low-resource dialects, especially in the Bangladeshi context. First, expanding and refining dialectal corpora is crucial. Ben-10 provides an excellent starting point, but additional data from under-represented districts and social groups, collected with consistent transcription and annotation protocols, would enable more robust training and evaluation [7]. Even a smaller, well-curated corpus of Bangladeshi local dialects can be highly valuable if it is designed with clear coverage goals and reliable metadata.

Second, systematic comparison and combination of model families remains important. Building on Ridoy *et al.* and Biswas *et al.*, one can investigate hybrid approaches that use Wav2Vec-BERT or WavLM encoders but borrow decoding strategies from Whisper-style models, potentially augmented with external language models trained on dialect-rich text [8, 3, 5, 6]. Parameter-efficient fine-tuning methods may facilitate maintaining both a shared core and dialect-specific variants without excessive compute.

Third, deploying these systems in realistic Bangladeshi applications will require continued attention to noise robustness and code-switching. The WavLM results show that noise-aware training substantially improves performance at low SNR; extending this work to include Bangla–English code-switched speech and telephone-channel distortions would bring the models closer to real deployment scenarios such as call centres or mobile voice interfaces [6, 10]. Fourth, future evaluations should incorporate fairness-oriented metrics, reporting not only overall WER but also stratified scores by dialect, gender, age and region. Aligning Bengali dialect ASR research with broader work on low-resource and inclusive ASR would help ensure that technological advances benefit speakers across Bangladesh rather than amplifying existing imbalances [11].

In summary, the current state of research suggests that Bengali dialect ASR is technologically feasible but far from solved. The combination of self-supervised encoders like Wav2Vec-BERT and WavLM, dialect-aware multi-stage adaptation strategies and dedicated resources such as Ben-10 and automatically crawled corpora provides a solid foundation. Building on this foundation with Bangladeshi local dialect data, careful model evaluation and fairness-aware design will be key to realising practical, inclusive ASR applications for Bengali speakers in diverse regions and social contexts.

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